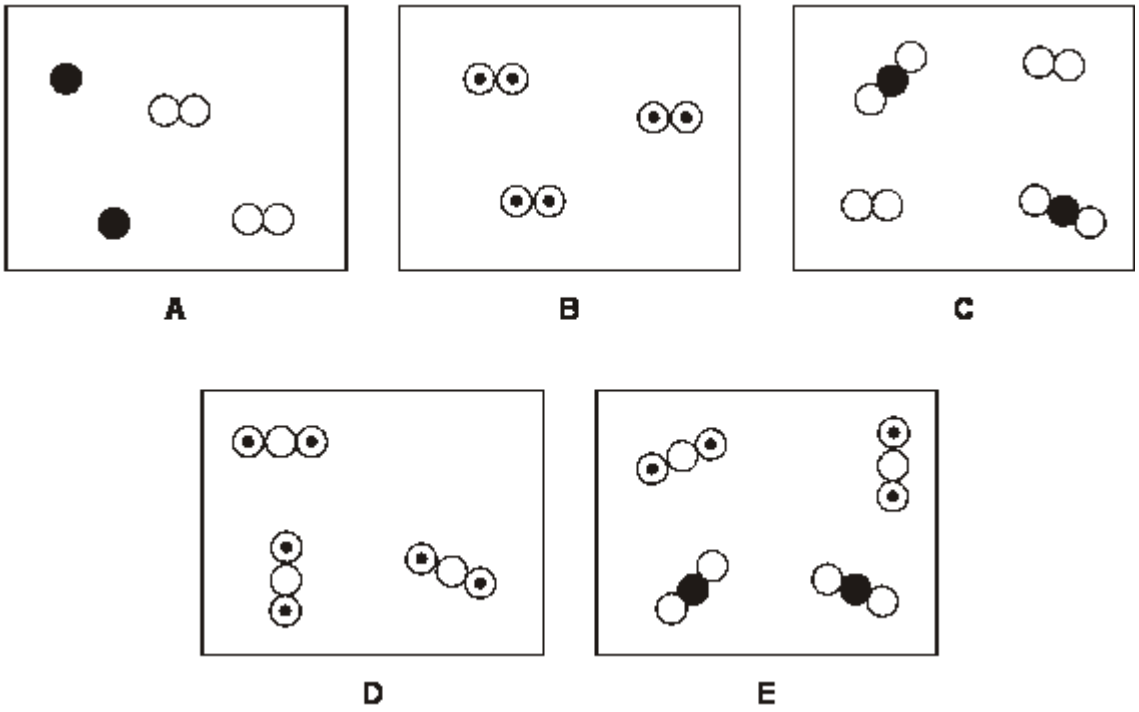
**Q1.**

In the 19th Century, a scientist called John Dalton used symbols to represent atoms. The symbols he used for atoms of three different elements are shown below.



The diagrams below show different combinations of these atoms.




(a) (i) Give the letter of the diagram which shows a mixture of **two** elements.

.....

1 mark

(ii) Give the letter of the diagram which shows a mixture of **two** compounds.

.....

1 mark

(iii) Give the letter of the diagram which shows a mixture of an element and a compound.

.....

1 mark

(b) Give **one** difference between a compound and a mixture.

.....

.....

.....

1 mark

(c) (i) Suggest a name and formula for the substance represented in diagram B.

name .....

formula .....

1 mark

(ii) Suggest a name and formula for the substance represented in diagram D.

name .....

formula .....

1 mark
maximum 6 marks

## Q1.

(a) (i) A

**1 (L7)**

(ii) E

**1 (L7)**

(iii) C

**1 (L7)**

(b) any **one** from

- a compound contains two **or** more elements chemically combined **or** bonded

  *accept 'a compound contains different elements **or** atoms of different types joined together'*

  *do **not** accept 'a compound contains two **or** more atoms joined together'*

- a mixture contains two **or** more elements **or** compounds **or** substances not chemically combined **or** bonded

  *accept 'a mixture can be separated by physical means'*

  *'a mixture can be separated' is insufficient*

**1 (L7)**

(c) (i) any **one** from

- *name:* oxygen *formula:* $O_2$
- *name:* hydrogen *formula:* $H_2$
- *name:* nitrogen *formula:* $N_2$
- *name:* chlorine *formula:* $Cl_2$

  *accept any other element which exists as diatomic molecules in the gaseous state*

  ***both** the name and formula are required for the mark*

  *do **not** accept 'O' **or** 'H' **or** 'N' **or** 'Cl'*

**1 (L7)**

(ii) any **one** from

- *name:* carbon dioxide *formula:* $CO_2$
- *name:* water *formula:* $H_2O$
- *name:* sulphur dioxide *formula:* $SO_2$
- *name:* nitrogen dioxide **or** nitrogen oxide *formula:* $NO_2$

  *accept any compound with the formula $XY_2$, eg magnesium chloride $MgCl_2$*

  ***both** the name and formula are required for the mark*

**1 (L7)**

**[6]**